

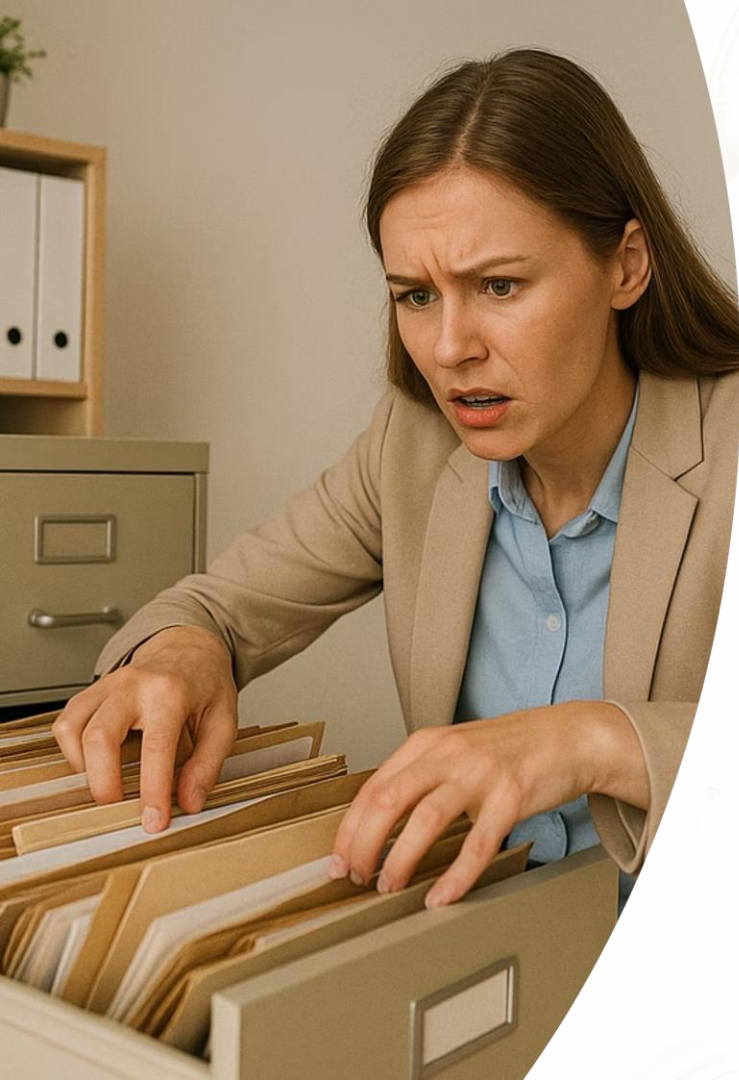


ITC Hotels — Revenue Optimization & Business Intelligence

This project involves the design and development of an interactive, multi-page Power BI dashboard for ITC Hotels, a luxury hotel chain operating multiple properties across India. The dashboard provides stakeholders with real-time, data-driven insights across four critical business dimensions: financial performance, occupancy analysis, room category bookings, and cancellation behaviour — enabling informed decision-making to maximize revenue and operational efficiency.

Presented By

Prakash Chawda



The Problem

ITC Hotels operates across multiple cities with diverse room categories and varying occupancy rates. Despite generating substantial revenue, the organization lacked a centralized, visual analytics platform — forcing management to rely on fragmented reports that delayed decision-making and obscured optimization opportunities.

Financial Blind Spots

No real-time monitoring of revenue across hotels and room types

Occupancy Gaps

No clear view of occupancy trends across seasons, day types, and hotel categories

Revenue Leakage

Cancellation patterns and lost revenue were unquantified and untracked

Room Category Performance

Which room categories drive the most revenue and bookings remained unclear

Project Objectives

Six primary objectives guided the design and development of this Power BI dashboard, covering all key performance areas for ITC Hotels stakeholders.

01

Comprehensive Dashboard

Build an interactive Power BI dashboard covering all key performance areas

03

Occupancy Analysis

Analyse occupancy patterns by hotel, room category, city, and day type

05

Cancellation Quantification

Quantify cancellation rates and lost revenue to identify corrective actions

02

Revenue KPIs

Track Total Revenue, Cumulative Revenue, MoM/WoW Growth, ADR, and RevPAR

04

Booking Behaviour

Evaluate room category and booking behaviour including ALOS and lead time

06

Self-Service Filters

Provide drill-down filters (city, category, room type, month, day) for stakeholder self-service

Dataset Overview

The dataset contains hotel booking records for ITC Hotels spanning **May–July 2022**, covering 7 properties across multiple Indian cities with over 233,000 customer records.

Hotels Covered	7 Properties
Time Period	May – July 2022
Total Customers	233,000+
Room Categories	RT1, RT2, RT3, RT4 (Elite, Premium, Presidential, Standard)
Cities	Agra, Bangalore, Delhi, Goa, Hyderabad, and more
Hotel Categories	Business & Luxury
Booking Types	Weekday & Weekend

Tools & Techniques

Why Power BI?

- Native **DAX** support for complex KPI calculations — MoM Growth, WoW Growth, RevPAR, ADR, Moving Averages, Cumulative measures
- Interactive **cross-filtering** — selecting a city, hotel, or month automatically updates all visuals
- Multi-page dashboard with intuitive navigation buttons
- Rich visualization library — bar charts, line charts, treemaps, KPI cards, matrices
- Slicer panels for non-technical stakeholder self-service
- Efficient processing of **233K+ records** via Power Query and in-memory columnar storage
- Professional publishing via **Power BI Service**

Other Techniques Used

Power Query (M Language)

Data cleaning, type casting, date parsing, and table transformations

DAX Measures

Custom calculations for all KPIs

Data Modelling

Star schema relationships between booking, hotel, room, and date dimension tables

Conditional Formatting

Visual cues for performance thresholds in matrix tables

Key DAX Formulas

Twelve+ custom DAX measures were developed to power all KPIs across the dashboard, including time-intelligence functions for MoM, WoW, moving averages, and cumulative calculations.

KPI	DAX Formula	Purpose
Total Revenue	SUM(Bookings[Revenue])	Sum of all confirmed booking revenue
Cumulative Revenue	CALCULATE([Total Revenue], FILTER(ALL(Date), Date[Date] ≤ MAX(Date[Date])))	Running total of revenue over time
MoM Growth	(Current Month – Previous Month) / Previous Month	Month-over-month revenue growth rate
WoW Growth	(Current Week – Previous Week) / Previous Week	Week-over-week revenue growth rate
Moving Average	AVERAGEX(DATESINPERIOD(...), [Total Revenue])	Smoothed revenue trend
ADR	DIVIDE([Total Revenue], [Room Nights Sold])	Revenue per occupied room per night
RevPAR	DIVIDE([Total Revenue], [Total Available Room Nights])	Revenue per available room
Occupancy Rate	DIVIDE([Arrived Customers], [Total Customers])	% of available rooms occupied
Cancellation Rate	DIVIDE([Cancelled Bookings], [Total Bookings])	% of bookings cancelled
Lost Revenue	SUM(Cancelled[Revenue])	Total revenue lost from cancellations
ALOS	DIVIDE([Total Stay Nights], [Total Bookings])	Average length of stay
Avg Lead Time	DIVIDE([Sum of Lead Time], [Total Bookings])	Avg days between booking and check-in

Method:

1

Step 1: Data Loading

Obtained the ITC Hotels dataset. Loaded CSV files into Power BI Desktop via Power Query Editor.

2

Step 2: Data Cleaning

Removed nulls and duplicates, standardized date formats, created calculated columns (Stay Duration, Lead Time, Day Type: Weekday/Weekend, Month-Year labels), and categorized room types.

3

Step 3: Data Modelling

Established relationships between fact table (Bookings) and dimension tables (Hotels, Rooms, Dates). Built a star schema to optimize query performance.

4

Step 4: DAX Development

Wrote all 12+ DAX measures, including time-intelligence functions for MoM, WoW, moving averages, and cumulative calculations.

5

Step 5: Dashboard Design

Built four interactive report pages as per project requirements. Applied ITC Hotels branding (orange theme, gold accents). Added page-level slicers and navigation buttons.

6

Step 6: Validation & Publishing

Cross-verified all KPIs against raw data. Validated formulas, checked slicer interactions, and exported the final report as a PDF for submission.

Total Revenue

2bn

Cum. Revenue

2bn

MOM Growth

41.27%

WOW Growth

4.43%

Moving Avg

21.01M

Avg Daily Rate

12.70K

RevPAR

7.35K

Financial Overview &
Revenue Performance

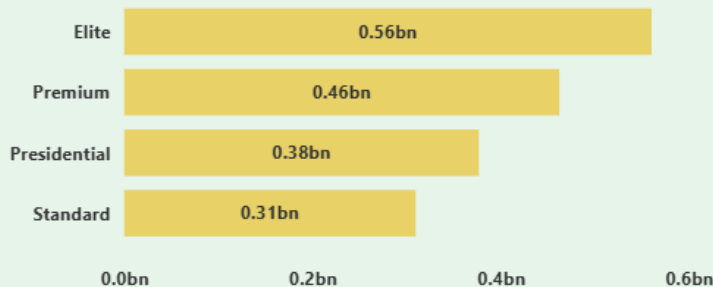
category ▾

☐ Business☐ Luxury

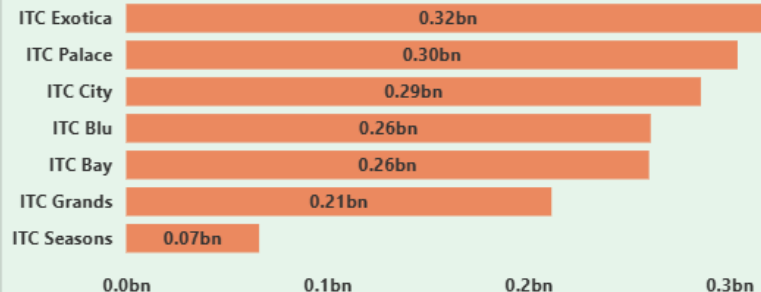
city ▾

☐ Agra☐ Bangalore☐ Delhi☐ Goa

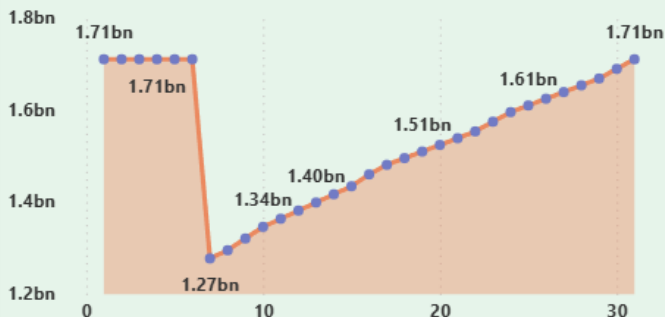
Room Class Wise Revenue



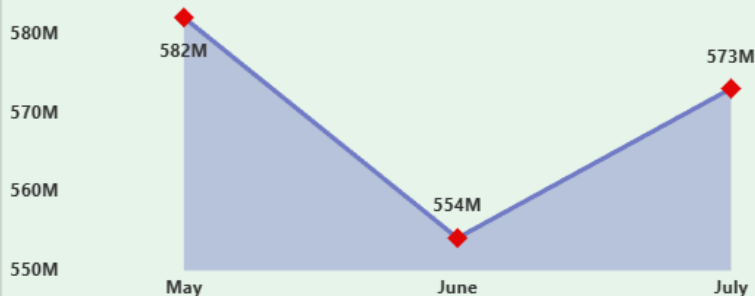
Hotel Wise Revenue



Day Wise Cumulative Revenue



Month Wise Revenue



Financial Performance

₹2B

Total Revenue

Strong aggregate performance across 3 months

41.27%

MoM Growth

Significant month-over-month expansion

4.43%

WoW Growth

Steady weekly revenue increase

₹12,700

ADR

Premium pricing aligned with luxury positioning

₹7,350

RevPAR

Effective room yield given ~58% occupancy

₹21.01M

Moving Average

Stable daily revenue baseline



Top Hotel

ITC Exotica — ₹0.32B, followed by ITC Palace (₹0.30B) and ITC City (₹0.29B)



Needs Focus

ITC Seasons — ₹0.07B (lowest-performing, requires strategic attention)



Top Room Category

Elite rooms — ₹0.56B; Standard rooms generated least at ₹0.31B



Monthly Trend

May (₹582M) > July (₹573M) > June (₹554M) — June dip signals seasonal softness

Total
Customers
233K

Arrived
Customers
135K

MOM Growth
-0.36%

WOW Growth
-0.96%

Cum.
Customers
135K

Occupancy
Rate
57.87%



Occupancy &
Capacity Analysis

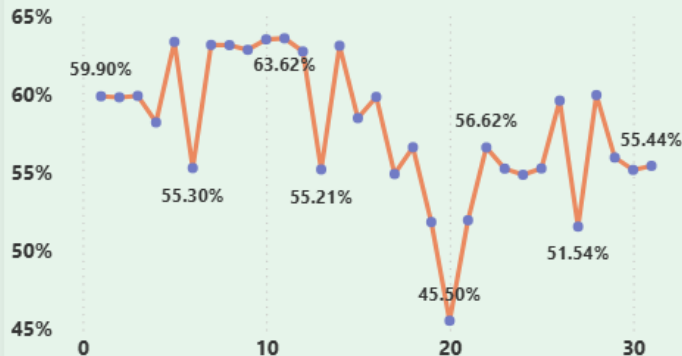
category ▾

- ☐ Business
- ☐ Luxury

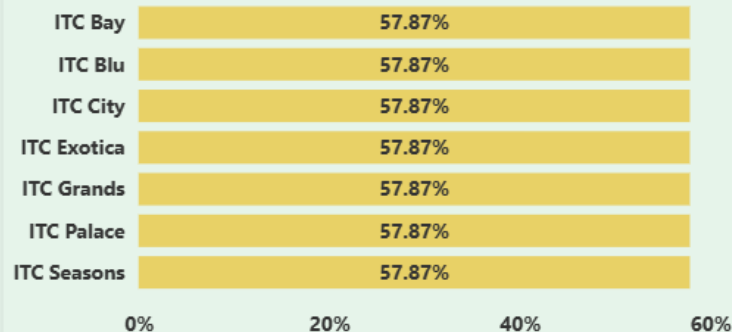
room_... ▾

- ☐ Elite
- ☐ Premium
- ☐ President...
- ☐ Standard

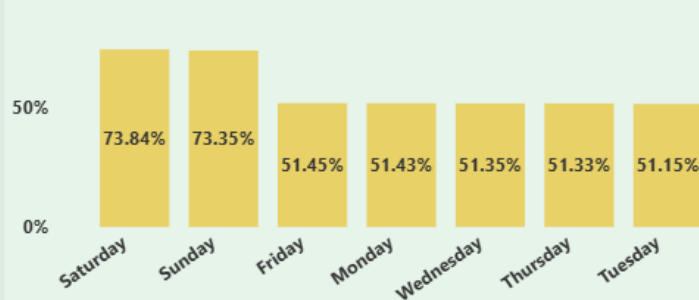
Day Wise Occupancy Rate



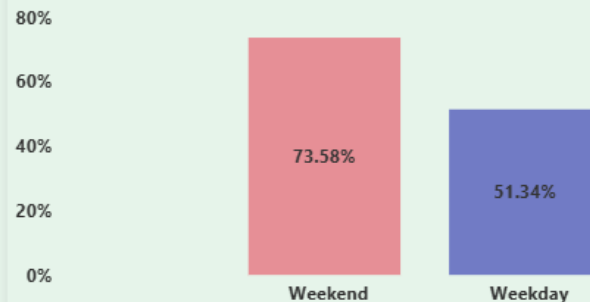
Hotel Wise Occupancy Rate



Room Class Wise Occupancy Rate



Day wise Occupancy Rate



Occupancy & Capacity Analysis

Core Occupancy KPIs

Total Customers: 233,000 bookings attempted

Arrived Customers: 135,000 confirmed check-ins

Occupancy Rate: 57.87% — ~42% capacity still underutilized

MoM Occupancy Growth: -0.36% (slight decline)

WoW Occupancy Growth: -0.96% (mild softness)

Key Observations

- All 7 hotels show **identical overall occupancy (57.87%)** — indicating consistent pricing and booking spread
- Date-wise analysis reveals **high volatility**: lowest recorded 45.50%, peak at 63.62%
- Business and Luxury categories show **equal occupancy (57.87%)** — no category dominance



Total Revenue

2bn

Avg Len Of

Stay

2.37

MOM Growth

41.27%

WOW Growth

4.43%

Average Lead

Time

3.71

Room Category
Performance &
Booking Insights

Month

☐ April

☐ August

☐ July

☐ June

☐ May

city

☐ Agra

☐ Bangalore

☐ Delhi

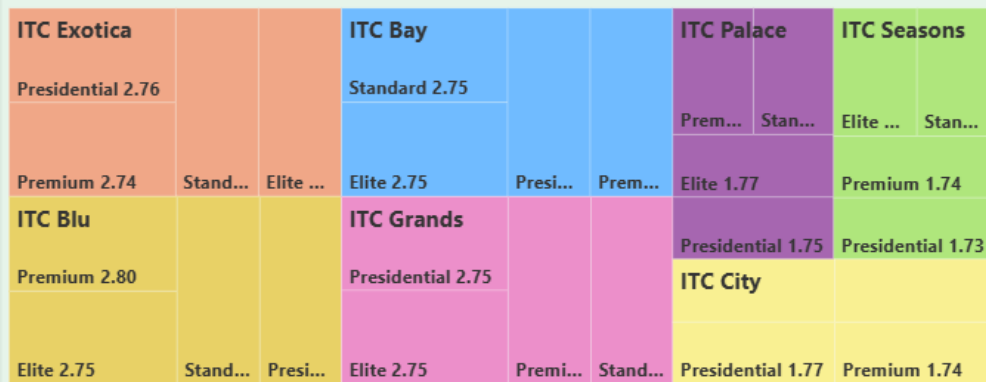
☐ Goa

☐ Hyderabad

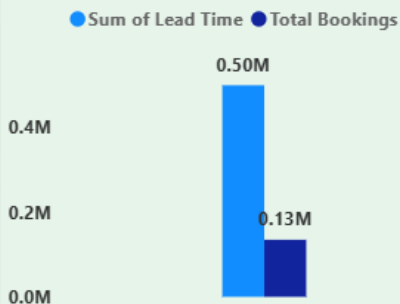
Room Class Wise Revenue



Hotel And Room Class Wise Revenue



Lead Time V/S Total Booking



Day Wise Cumulative Revenue



Stay Duration Wise Revenue



Room Category Performance & Booking Insights

Elite and Premium rooms command the longest stays (ALOS 2.74–2.80 nights), while ITC Exotica leads in Presidential room performance (ALOS: 2.76). ITC Seasons records the lowest ALOS at ~1.75 nights. Revenue is heavily concentrated at 1–2 night bookings, with sharp drops beyond 2 nights.

2.37

Avg Length of Stay

Nights — short-stay dominant market

3.71

Avg Lead Time

Days — last-minute booking pattern

0.50M

Lead Time Volume

vs 0.13M bookings — multiple look-ups

₹0.81B

Peak Stay Revenue

2-night stays dominate revenue

Lost Revenue
299M

Total Bookings
135K

MOM Growth
40.02%

WOW Growth
4.13%

Cancellation
Rate
24.83%

Cancelled
Bookings
33K

Cum.
Cancellation
33K



Cancellations & Lost Revenue

category

☐ Business

☐ Luxury

city

☐ Agra

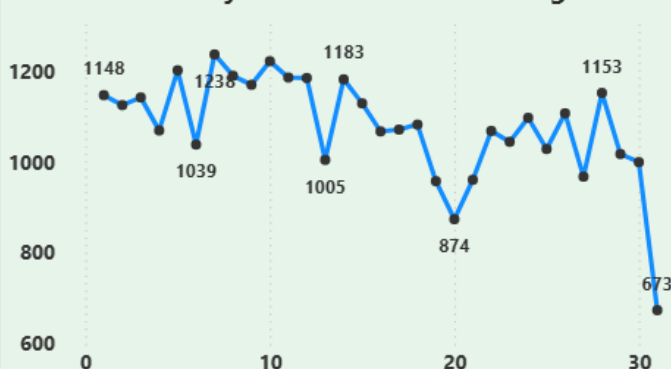
☐ Bangalore

☐ Delhi

☐ Goa

☐ Hyderabad

Day Wise Cancelled Booking



Day

☐ Friday

☐ Monday

☐ Saturday

☐ Sunday

☐ Thursday

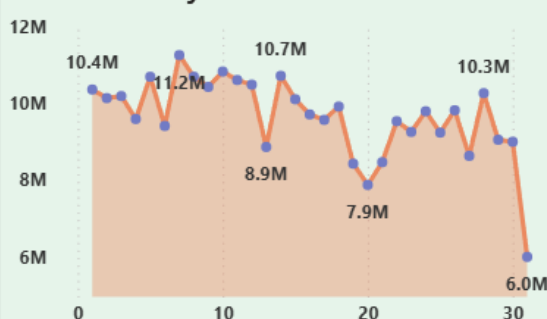
☐ Tuesday

☐ Wednesday

Room Class Wise Cancellation Rate

Month	Elite	Premium	Presidential	Standard
July	24.33%	24.99%	24.24%	24.30%
June	25.61%	24.85%	24.51%	24.86%
May	24.97%	24.81%	24.56%	25.20%
Total	24.96%	24.88%	24.44%	24.79%

Day Wise Revenue Lost



Room Class Wise Cancellation Rate

Elite	24.96%
Premium	24.88%
Standard	24.79%
Presidential	24.44%

Hotel Wise Cancellation Rate

ITC Palace	25.18%
ITC Grands	25.08%
ITC City	24.92%
ITC Bay	24.84%
ITC Seasons	24.79%
ITC Blu	24.65%
ITC Exotica	24.37%

Cancellations & Lost Revenue: A ₹299 Million Leak

Nearly 1 in 4 bookings is cancelled, representing ₹299 Million in preventable revenue loss — approximately 15% of total portfolio revenue. The cancellation rate is accelerating at 40.02% MoM, signaling an urgent need for intervention.

24.83%

Cancellation Rate

Nearly 1 in 4 bookings cancelled

₹299M

Lost Revenue

~15% of total revenue lost to cancellations

33,000

Cancelled Bookings

Significant volume out of 135K arrived bookings

40.02%

MoM Growth

Cancellations growing rapidly month-over-month

⚠ Week-over-week cancellation growth of 4.13% confirms the trend is accelerating — not a one-time anomaly.

What Was Accomplished

1

Interactive Power BI Dashboard

Delivered a fully functional 4-page dashboard with 5-page navigation, enabling self-service analytics for non-technical stakeholders.

2

12+ Custom DAX Measures

Covering all financial, occupancy, and booking KPIs with drill-down analysis via slicers for City, Hotel Category, Room Type, Month, and Day Type.

3

₹299M Revenue Leakage Identified

Quantified preventable lost revenue from cancellations — directly addressable through policy and retention interventions.

4

Key Tension Uncovered

Elite rooms are both the top revenue generator and most-cancelled category — a critical operational tension requiring targeted management.

5

RevPAR Gap Confirmed

RevPAR of ₹7,350 vs ADR of ₹12,700 confirms significant underutilized capacity across the portfolio.

Limitations of This Analysis

Data Scope Constraints

- Dataset covers only 3 months (May–July 2022) — insufficient for annual seasonality or year-over-year comparison
- Occupancy rates appear uniform (57.87%) across all hotels — possible data aggregation issue; hotel-specific variation may be masked
- No customer demographic data — unable to segment by age, nationality, or loyalty tier

Analytical Constraints

- External factors (events, weather, competitor pricing) not included — limits causal analysis
- Real-time data refresh not implemented — dashboard reflects a static historical snapshot
- Average lead time of 3.71 days limits predictive forecasting accuracy

 These limitations define the roadmap for Phase 2 enhancements.

Future Opportunities



ITC HOTELS
RESPONSIBLE LUXURY



Live Data Integration

Connect Power BI Service to hotel PMS via scheduled refresh for real-time performance monitoring.



Predictive Analytics

Embed Python or R visuals to forecast revenue and occupancy using lead time and seasonal patterns.



Customer Segmentation

Build a dedicated page using loyalty data, stay frequency, and spend analysis to personalize offers.



Dynamic Pricing Engine

Implement pricing recommendations based on occupancy thresholds and lead time signals.



Cancellation Alert System

Deploy Power BI data alerts to flag cancellation rate spikes before they become revenue crises.



12-Month Dataset & Benchmarking

Expand to 12+ months for seasonal detection and integrate STR report benchmarks for competitive context.

Recommendations

Reduce Cancellations with Booking Policy

Implement non-refundable or partial-refund policies for bookings with <3 days lead time to recover the ₹299M in lost revenue and counter the 24.83% cancellation rate.

Revitalize ITC Seasons

Target ITC Seasons with promotional campaigns — it generates only ₹0.07B vs ₹0.32B for ITC Exotica, a 4.5x gap representing significant untapped potential.

Upsell to Elite Rooms in June

Invest in upselling Standard room guests to Elite category during June — historically the weakest month with a ₹554M revenue dip — to lift overall performance.

Boost Weekend Occupancy

Introduce weekend packages to address the 45.50% occupancy trough on low-demand days and raise RevPAR above ₹7,350.

Design 2-Night Experience Bundles

Capitalize on short-stay dominance by packaging 2-night experiences to increase ALOS from 2.37 to 3+ nights and grow per-booking revenue.

Conclusion

This project demonstrates the power of **Microsoft Power BI** as a business intelligence tool for the hospitality sector. By transforming raw booking data into a structured, interactive dashboard, it surfaced critical insights that would otherwise remain hidden in spreadsheets.

₹2B Revenue, 41% MoM Growth

Strong top-line performance with significant momentum — but ₹299M in preventable cancellation losses must be addressed.

24.83% Cancellation Rate

Growing at 40% MoM — an urgent signal requiring policy intervention, retention incentives, and real-time alerting.

Scalable BI Foundation

Power BI's DAX engine, visualization layer, and interactive slicers provide ITC Hotels leadership with a scalable platform for ongoing performance monitoring and data-driven planning.

- ✓ The resulting dashboard provides ITC Hotels leadership with a scalable foundation for ongoing performance monitoring, operational improvement, and strategic planning — turning data into competitive advantage.

Note : This Whole Project(Dashboard) available on LinkedIn and GitHub . These Link are Clickable